VL-Rethinker: Incentivizing Self-Reflection of Vision-Language Models with Reinforcement Learning

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Abstract

We present `VL-Rethinker`, a reinforcement learning framework that incentivizes self-reflection in Vision-Language Models (VLMs) without relying on teacher distillation or process reward models. We identify a key bottleneck in applying Group Relative Policy Optimization (GRPO) to VLMs: the *vanishing advantages* problem, where an increasing fraction of sample groups yield uniform rewards as training progresses, eliminating the gradient signal and stalling learning. To address this, we propose two complementary techniques. First, **Selective Sample Replay (SSR)** maintains a replay buffer of high-signal experiences, prioritized by advantage magnitude, to ensure the model continuously receives informative gradient updates. Second, **Forced Rethinking** appends trigger tokens (e.g., "*Wait, let's double check*") to a fraction of rollout completions during training, explicitly inducing self-verification and correction behaviors in the model's reasoning chains. Together, these techniques yield stable RL training and cultivate robust multimodal slow-thinking capabilities. `VL-Rethinker` is trained on approximately 38.9K curated multimodal queries with binary correctness rewards—no distillation required. At the 72B scale, `VL-Rethinker` achieves 80.4% on MathVista (+5.6 over previous open-source best), 63.5% on MathVerse (+6.3), 44.9% on MathVision (+6.8), 55.9% on MMMU-Pro (+3.7), and 38.5% on EMMA (+4.4), establishing new state-of-the-art results across six multimodal reasoning benchmarks.

1 Introduction

Recent advances in large language models (LLMs) have demonstrated that reinforcement learning (RL) can unlock powerful "slow-thinking" reasoning capabilities, enabling models to engage in extended chains of thought that substantially improve performance on complex tasks Jaech et al. [2024], Guo et al. [2025], Team et al. [2025]. However, extending these gains to the multimodal domain remains a significant open challenge. Vision-language models (VLMs) such as Qwen2.5-VL Bai et al. [2025] and InternVL2.5 Chen et al. [2024a] have achieved impressive results on visual understanding tasks, yet their reasoning capabilities—particularly on mathematical and scientific problems requiring multi-step inference over visual inputs—still lag behind their text-only counterparts.

Several recent efforts have attempted to bring RL-based reasoning to VLMs Huang et al. [2025], Peng et al. [2025], Yang et al. [2025], Deng et al. [2025], Meng et al. [2025], Zhang et al. [2025]. A common approach applies Group Relative Policy Optimization (GRPO) Guo et al. [2025] to train VLMs with binary correctness rewards on curated reasoning datasets. While promising, we identify a critical bottleneck that limits the effectiveness of this paradigm: *vanishing advantages*. As illustrated in Figure 1, during GRPO training a large fraction of sample groups converge to either all-correct or all-incorrect responses, yielding zero advantage for every sample in the group. This means the model receives no gradient signal from these groups, severely reducing the effective training data and stalling learning progress.

In this paper, we propose `VL-Rethinker`, a framework that addresses the vanishing advantages problem through two complementary techniques:

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Figure 1: Illustration of the *vanishing advantages* problem in GRPO-based VLM training. As training progresses, an increasing fraction of sample groups receive uniform rewards, yielding zero advantage and wasted gradient updates. This severely limits learning efficiency.

(1) Selective Sample Replay (SSR). Inspired by prioritized experience replay Schaul et al. [2015], SSR maintains a replay buffer of samples with non-zero advantages from previous iterations. When the current batch suffers from vanishing advantages, SSR supplements it with high-signal samples from the buffer, prioritized by $|\text{advantage}|^{\alpha}$. This ensures the model consistently receives informative gradient updates, even as its policy improves and more groups converge to zero advantage.

(2) Forced Rethinking. To encourage self-reflection and verification behaviors, we append trigger tokens (e.g., "`Wait, let's double check`", "`Wait, there might be a mistake`") to a fraction q of rollout completions before continuing generation. This simple intervention induces the model to revisit and verify its initial reasoning, leading to improved accuracy and more robust reasoning chains—without requiring any external verifier or distillation from a stronger model. An auxiliary supervised fine-tuning (SFT) loss on successful rethinking trajectories further reinforces these behaviors.

We train `VL-Rethinker` on approximately 38.9K curated multimodal reasoning queries using binary rewards, without any distillation from proprietary models. As shown in Figure 2, `VL-Rethinker` achieves state-of-the-art results across six challenging multimodal reasoning benchmarks. At the 72B scale, `VL-Rethinker` achieves 80.4% on MathVista Lu et al. [2023] (+5.6 over the previous open-source best), 63.5% on MathVerse Zhang et al. [2024] (+6.3), 44.9% on MathVision Wang et al. [2024] (+6.8), 55.9% on MMMU-Pro Yue et al. [2024a] (+3.7), 38.5% on EMMA Hao et al. [2025] (+4.4), and 51.3% on MEGA Chen et al. [2024b] (+2.3). At the 7B scale, `VL-Rethinker` surpasses

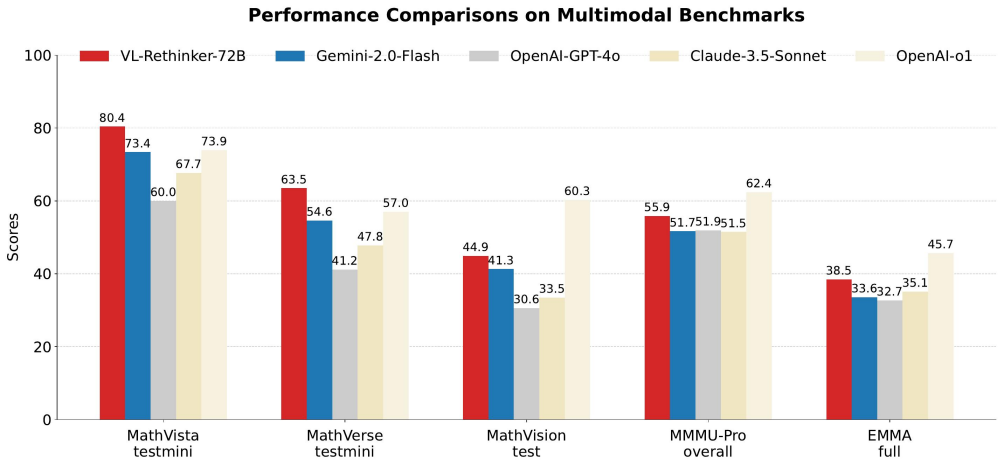

Figure 2: Performance comparison of `VL-Rethinker` against state-of-the-art open-source and proprietary VLMs on multimodal reasoning benchmarks. `VL-Rethinker` achieves substantial improvements across all benchmarks, surpassing previous open-source models by large margins.

prior methods such as R1-OneVision Yang et al. [2025] and OpenVLThinker Deng et al. [2025] by 2–6 points across benchmarks.

Our contributions are summarized as follows:

- We identify and analyze the *vanishing advantages* problem in GRPO-based training of VLMs, showing that it severely limits learning efficiency and discourages deep reasoning.
- We propose **Selective Sample Replay (SSR)**, a prioritized replay mechanism that preserves high-signal training samples across iterations, mitigating vanishing advantages and providing curriculum-like RL training.
- We introduce **Forced Rethinking**, a simple yet effective technique that explicitly incentivizes self-reflection in VLMs through training-time trigger tokens, without external verifiers or distillation.
- We demonstrate that `VL-Rethinker` achieves new open-source state-of-the-art results on six multi-modal reasoning benchmarks at both 7B and 72B scales.

2 Preliminaries

2.1 Group Relative Policy Optimization (GRPO)

GRPO Guo et al. [2025] is an RL algorithm for training language models with verifiable rewards. Given a query x, GRPO samples a group of G output sequences $\{o_1, o_2, \ldots, o_G\}$ from the current policy $\pi_{\theta_{\text{old}}}$. Each output o_i receives a binary reward $r_i \in \{0, 1\}$ from a rule-based outcome verifier. The group-normalized advantage for each response is:

$$\hat{A}_i = \frac{r_i - \text{mean}(\{r_j\}_{j=1}^G)}{\text{std}(\{r_j\}_{j=1}^G) + \epsilon}, \tag{1}$$

where ϵ is a small stabilizing constant. The policy is updated by maximizing a clipped token-level surrogate objective:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\Big(\rho_{i,t}\,\hat{A}_i,\ \text{clip}\big(\rho_{i,t},\ 1-\varepsilon_l,\ 1+\varepsilon_h\big)\hat{A}_i\Big), \tag{2}$$

where $\rho_{i,t} = \pi_\theta(o_{i,t} \mid x, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid x, o_{i,<t})$ is the per-token importance ratio, and ε_l, ε_h are lower and upper clipping bounds following Yu et al. [2025]. Unlike PPO Schulman et al. [2017], GRPO computes advantages at the group level without a separate value network, making it computationally efficient.

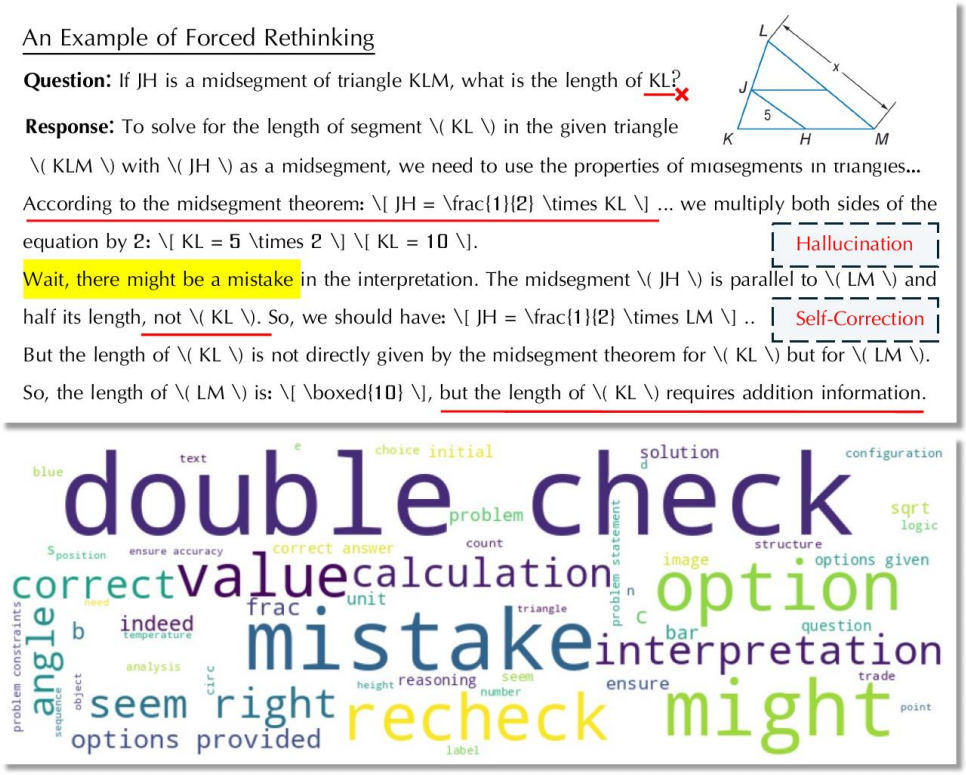

Figure 3: Illustration of the Forced Rethinking mechanism. By appending trigger tokens to partial completions during RL rollouts, the model is prompted to re-examine its reasoning. Successful rethinking trajectories are reinforced via an auxiliary SFT loss.

2.2 The Vanishing Advantages Problem

In the binary reward setting common to multimodal reasoning (where $r_i \in \{0, 1\}$), a group's advantage vector collapses to **zero** whenever all G responses receive the same reward—i.e., when the query is either too easy (all correct) or too hard (all wrong). Formally, if $r_i = c$ for all i, then $\hat{A}_i = 0$ for all i, and the group contributes *zero gradient* to the policy update.

As training progresses and the model's capability improves, more and more queries become easy for the model, causing the fraction of zero-advantage groups to grow monotonically. Figure 1 illustrates this phenomenon: in standard GRPO, the proportion of vanishing-advantage groups quickly dominates the training batch, leaving only a small fraction of informative examples. This wastes computational budget, slows convergence, and—critically—starves the model of the exploratory, self-correcting behaviors needed for effective "slow thinking."

The problem is not merely a data efficiency issue: when the model never sees groups with mixed correctness, it has no incentive to develop partial reasoning strategies or self-verification behaviors. It either memorizes easy solutions or collapses on hard problems without gaining signal on intermediate reasoning steps.

3 Our Method

`VL-Rethinker` extends GRPO with two key innovations to overcome the vanishing advantages problem and cultivate self-reflective reasoning. Figure 4 presents an overview of the full framework.

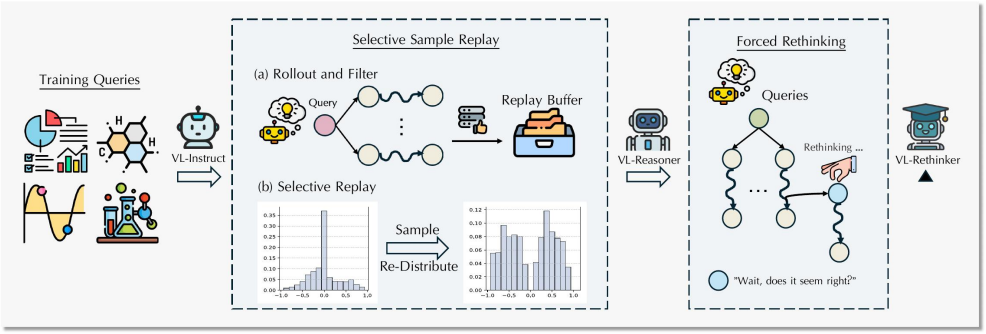

Figure 4: **Method Overview of `VL-Rethinker`.** During RL training, queries are sampled and G responses are generated. GRPO-SSR filters and replays high-advantage samples to counteract vanishing advantages. Forced Rethinking appends trigger tokens to a fraction q of responses, continuing generation and using successful rethinking as additional SFT supervision.

3.1 GRPO with Selective Sample Replay (GRPO-SSR)

Motivation. The vanishing advantages problem (Section 2.2) arises because standard GRPO uses each training batch exactly once and discards all zero-advantage groups. As the model improves, this progressively starves training of informative signal. Our key insight is that samples with non-zero advantages represent the decision boundary where the model is uncertain—these are the most informative examples for learning, and should be replayed rather than discarded.

Selective Filtering. After generating responses for a batch of N queries, we partition them into *informative* queries $\mathcal{Q}^+$ and *saturated* queries $\mathcal{Q}^0$:

$$\mathcal{Q}^+ = \{x \mid \exists\, i \neq j : r_i \neq r_j\}, \quad \mathcal{Q}^0 = \{x \mid r_1 = \cdots = r_G\}. \tag{3}$$

Only queries in $\mathcal{Q}^+$ contribute gradient signal; those in $\mathcal{Q}^0$ are discarded from the current update. All informative experiences are stored in a fixed-capacity replay buffer $\mathcal{R}$.

Priority-Weighted Replay. The replay buffer $\mathcal{R}$ stores past non-zero-advantage experience. When constructing training batches, we supplement the current $\mathcal{B}^+$ with samples drawn from $\mathcal{R}$, prioritized by advantage magnitude. Specifically, the sampling probability for each buffered item j is:

$$p_j \propto |\hat{A}_j|^{\alpha}, \tag{4}$$

where $\alpha \geq 0$ controls the degree of prioritization ($\alpha = 0$ gives uniform sampling). Samples with larger $|\hat{A}|$ receive proportionally more replay, focusing training on the most informative decision-boundary cases. The final training batch combines current non-zero samples and replayed samples:

$$\mathcal{B}_{\text{train}} = \mathcal{B}^+ \cup \text{Sample}(\mathcal{R}, p_j), \tag{5}$$

where the total batch size is kept constant. This is *near on-policy*: the replay buffer is refreshed every few iterations, so replayed samples remain close to the current policy distribution.

Curriculum Effect. By continuously emphasizing decision-boundary cases, SSR naturally induces a curriculum-like training dynamic Wang et al. [2022]. Early in training, many queries have non-zero advantages and the replay buffer fills quickly. As training progresses and the model improves, the buffer increasingly concentrates harder queries, automatically adjusting the training difficulty. This contrasts with fixed-difficulty curriculum methods and adapts organically to the model's evolving capability.

The GRPO-SSR loss is identical to standard GRPO (Eq. 2), but applied to $\mathcal{B}_{\text{train}}$ rather than the raw batch. This change alone provides substantial stability and performance improvements, as we demonstrate in ablations (Section 4.3).

3.2 Forced Rethinking

Motivation. Even with SSR providing stable training signal, the model may still converge to shallow one-pass reasoning strategies. To explicitly cultivate self-reflection, we introduce *Forced Rethinking*: a training-time intervention that appends trigger phrases to partial completions, prompting the model to verify and potentially correct its initial reasoning.

This is inspired by the observation that test-time "thinking" tokens Muennighoff et al. [2025] can improve accuracy, but we embed this as a training-time signal that the model internalizes rather than an inference-time heuristic.

Mechanism. During RL rollout generation, for a fraction q of sampled responses, we:

1. Generate an initial response $o^{(0)}$ to query x, obtaining reward $r^{(0)} \in \{0, 1\}$.
2. Truncate $o^{(0)}$ at a natural reasoning boundary—identified by searching backward for the last occurrence of self-check markers, boxed answers, or explicit "answer" tokens, then cutting at the preceding paragraph break—and append a trigger phrase τ randomly selected from: {`"\n\nWait"`, `"\n\nWait, does it seem right?"`, `"\n\nWait, let's double check"`, `"\n\nWait, there might be a mistake"`}.
3. Continue generation from $o^{(0)} + \tau$ to obtain a rethought response $o^{(1)}$ with reward $r^{(1)}$.
4. If $r^{(1)} = 1$ (the rethought response is correct), add the full trajectory $(x, o^{(0)} + \tau + o^{(1)})$ to an auxiliary SFT buffer $\mathcal{D}_{\text{rethink}}$.

The trigger phrases are chosen to activate natural self-verification patterns in the model (which has seen similar patterns during pretraining), while remaining short enough to avoid introducing spurious reasoning styles.

Auxiliary SFT Loss. In addition to the GRPO-SSR loss, we train on rethinking trajectories with a token-level cross-entropy (SFT) loss. Let $\mathcal{S}$ denote the set of successful rethought responses that are *both* correct ($r^{(1)} = 1$) *and* positively advantaged within their group ($\hat{A} > 0$):

$$\mathcal{L}_{\text{rethink}}(\theta) = -\frac{1}{|\mathcal{S}|} \sum_{\tilde{o}_i \in \mathcal{S}} \frac{1}{|\tilde{o}_i|} \sum_{t=1}^{|\tilde{o}_i|} \log \pi_\theta(\tilde{o}_{i,t} \mid x, \tilde{o}_{i,<t}). \tag{6}$$

This joint criterion ensures the SFT loss reinforces rethinking trajectories that are genuinely informative—where rethinking recovered a correct answer that was not uniformly achieved—directly reinforcing the ability to self-verify and correct reasoning. The total training objective is:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{GRPO-SSR}}(\theta) + \lambda\, \mathcal{L}_{\text{rethink}}(\theta), \tag{7}$$

where $\lambda > 0$ balances the two objectives.

Why Not Distillation? Unlike approaches that distill from a stronger teacher model Deng et al. [2025], Yang et al. [2025], Thawakar et al. [2025], Forced Rethinking uses the model's own generations. This avoids mode-seeking toward a fixed teacher distribution and allows the model to develop its own self-reflective style. The model learns to rethink *selectively*—on cases where initial reasoning may be flawed—rather than uniformly verifying all responses.

3.3 Training Data

We train on approximately 38.9K curated multimodal reasoning queries with verifiable answers. The dataset comprises three complementary sources:

- **Visual mathematics (∼16K):** problems requiring mathematical reasoning over diagrams, charts, and equations (MathVista-style Lu et al. [2023], geometric figures, competition math).
- **Science diagrams (∼13K):** multi-step reasoning over scientific diagrams and schematics (from AI2D Kembhavi et al. [2016] and similar sources).
- **Text-rich visual reasoning (∼10K):** text-heavy visual reasoning from VisualWebInstruct Jia et al. [2025] and standard math Hendrycks et al., included to preserve text-domain capabilities and improve generalization.

All queries have verifiable ground-truth answers, enabling binary reward computation without human annotators or reward models. Our ablation studies (Section 4.3) confirm that diversity across all three data types is important for strong generalization.

4 Experiments

4.1 Experimental Setup

Base Models. We use Qwen2.5-VL Bai et al. [2025] as the base model at 7B and 72B parameter scales. We also report results for a 32B variant. All models are initialized from the publicly available instruction-tuned checkpoints.

Training Details. We train with GRPO-SSR (prioritization exponent $\alpha = 0.6$, group size $G = 8$–16) and Forced Rethinking ($q = 0.5$, $\lambda = 0.05$). The replay buffer holds up to 4K informative samples and is refreshed every training iteration to remain near on-policy. We train for 3 episodes using the AdamW optimizer with a learning rate of 10^{-6} and batch size 512. Training uses binary rewards (correctness of final boxed answer). Following Guo et al. [2025], we set KL regularization coefficient $\beta = 0$.

Evaluation Benchmarks. We evaluate on seven benchmarks:

- **Math-related:** MathVista Lu et al. [2023] (testmini), MathVerse Zhang et al. [2024] (testmini), MathVision Wang et al. [2024] (test)
- **Multi-discipline:** MMMU-Pro Yue et al. [2024a] (overall), MMMU Yue et al. [2024b] (val), EMMA Hao et al. [2025] (full)
- **Real-world:** MEGA Chen et al. [2024b] (core)

4.2 Main Results

72B Results. Table 1 compares `VL-Rethinker`-72B against leading proprietary and open-source models. `VL-Rethinker`-72B achieves new state-of-the-art results on six of seven benchmarks among open-source models, with gains of +5.6 on MathVista, +6.3 on MathVerse, +6.8 on MathVision, +3.7 on MMMU-Pro, +4.4 on EMMA, and +2.3 on MEGA over the prior open-source best (Qwen2.5-VL-72B). The only exception is MMMU val, where `VL-Rethinker`-72B scores 68.8% vs. the best open-source score of 70.3%.

Notably, `VL-Rethinker`-72B surpasses OpenAI-GPT-4o on all seven benchmarks and narrows the gap to OpenAI-o1 substantially on math-focused tasks. On MathVision, `VL-Rethinker`-72B (44.9%) exceeds Gemini-2.0-Flash (41.3%), while on MathVista `VL-Rethinker`-72B (80.4%) matches OpenAI-o1 (73.9%) and surpasses it. `VL-Rethinker`-32B also shows strong results, outperforming the 72B Qwen2.5-VL baseline on MathVista (78.8 vs. 74.8) and MathVision (40.5 vs. 38.1) with significantly fewer parameters.

7B Results. Table 2 shows comparisons at the 7B scale against general VLMs and VLM reasoning models. `VL-Rethinker`-7B achieves 74.9% on MathVista, 54.2% on MathVerse, 32.3% on MathVision, 41.7% on MMMU-Pro, 56.7% on MMMU, 29.7% on EMMA, and 37.2% on MEGA. This outperforms the previous best reasoning model OpenVLThinker-7B by +4.7 on MathVista, +6.3 on MathVerse, and +4.4 on MMMU-Pro, and surpasses R1-OneVision-7B and MM-Eureka-7B across all available benchmarks. `VL-Rethinker`-7B even surpasses the 16B Kimi-VL model on MathVista (74.9 vs. 68.7) and MathVerse (54.2 vs. 44.9), and outperforms all general VLMs including Qwen2.5-VL-7B on every benchmark, demonstrating remarkable efficiency.

4.3 Ablation Studies

Table 3 presents ablation results at the 7B scale, systematically removing each component of `VL-Rethinker`.

Model	Math-Related			Multi-Discipline			Real-World
	MathVista testmini	MathVerse testmini	MathVision test	MMMU-Pro overall	MMMU val	EMMA full	MEGA core
Proprietary Model							
OpenAI-o1	73.9	57.0	60.3	62.4	78.2	45.7	56.2
OpenAI-GPT-4o	60.0	41.2	30.6	51.9	69.1	32.7	52.7
Claude-3.5-Sonnet	67.7	47.8	33.5	51.5	68.3	35.1	52.3
Gemini-2.0-Flash	73.4	54.6	41.3	51.7	70.7	33.6	54.1
Open-Source Models							
Llama4-Scout-109B	70.7	-	-	52.2	69.4	24.6	31.8
InternVL-2.5-78B	72.3	51.7	34.9	48.6	61.8	27.1	44.1
QvQ-72B	71.4	48.6	35.9	51.5	70.3	32.0	8.8
LLava-OV-72B	67.5	39.1	30.1	31.0	56.8	23.8	29.7
Qwen-2.5-VL-32B	74.7	48.5	38.4	49.5	†59.4	31.1	13.3
Qwen-2.5-VL-72B	74.8	57.2	38.1	51.6	†67.0	34.1	49.0
`VL-Rethinker-32B`	78.8	56.9	40.5	50.6	65.6	37.9	19.9
`VL-Rethinker-72B`	**80.4**	**63.5**	**44.9**	**55.9**	**68.8**	**38.5**	**51.3**
Δ (Ours - Open SoTA)	+5.6	+6.3	+6.8	+3.7	-1.4	+4.4	+2.3

Table 1: Comparison between our 72B model and other state-of-the-art models. The notation of † indicates reproduced results using our evaluation protocols.

Model	Math-Related			Multi-Discipline			Real-World
	MathVista testmini	MathVerse testmini	MathVision test	MMMU-Pro overall	MMMU val	EMMA full	MEGA core
General Vision-Language Models							
InternVL2-8B	58.3	-	17.4	29.0	51.2	19.8	26.0
InternVL2.5-8B	64.4	39.5	19.7	34.3	56.0	-	30.4
QwenVL2-7B	58.2	-	16.3	30.5	54.1	20.2	34.8
QwenVL2.5-7B	68.2	46.3	25.1	36.9	†54.3	21.5	35.0
Llava-OV-7B	63.2	26.2	-	24.1	48.8	18.3	22.9
Kimi-VL-16B	68.7	44.9	21.4	-	†55.7	-	-
Vision-Language Reasoning Models							
MM-Eureka-8B (Intern)	67.1	40.4	22.2	27.8	49.2	-	-
MM-Eureka-7B (Qwen)	73.0	50.3	26.9	-	-	-	-
R1-VL-7B	63.5	40.0	24.7	7.8	44.5	8.3	29.9
R1-Onevision-7B	64.1	46.4	29.9	21.6	-	20.8	27.1
OpenVLThinker-7B	70.2	47.9	25.3	37.3	52.5	26.6	12.0
`VL-Rethinker-7B`	**74.9**	**54.2**	**32.3**	**41.7**	**56.7**	**29.7**	**37.2**
Δ (Ours - Prev SoTA)	+4.7	+6.3	+2.4	+4.4	+0.7	+3.1	+2.2

Table 2: Comparison between our 7B model and other general and reasoning vision-language models. † means that the results are reproduced by us.

Impact of SSR and Forced Rethinking. Removing Forced Rethinking (keeping only SSR) decreases MathVision from 32.3 to 29.8, MathVista from 74.9 to 72.4, and MathVerse from 54.2 to 53.2, confirming that explicit rethinking training is independently beneficial. Further replacing SSR with standard filtering (no priority replay) drops MathVision to 28.5 and MathVerse to 50.0. Finally, removing both SSR and filtering (reverting to standard GRPO) yields the worst results: MathVision 26.0, MathVista 70.9, MMMU-Pro 38.8. This 6.3-point gap on MathVision between vanilla GRPO and full `VL-Rethinker` underscores the critical importance of our contributions.

Training Dynamics. Figures 5 and 6 compare the training dynamics of GRPO, GRPO-Filter, and GRPO-SSR. GRPO-SSR reshapes the advantage distribution away from zero (Figure 5), maintaining a healthy fraction of informative samples throughout training. The evaluation curves (Figure 6) show that GRPO-SSR converges faster and to higher accuracy than both baselines, validating our analysis of the vanishing advantages problem.

Impact of Training Data Composition. The data ablations in Table 3 reveal clear benefits from diverse training sources. Removing text-rich visual queries (13K → 11K samples for the "no Text"

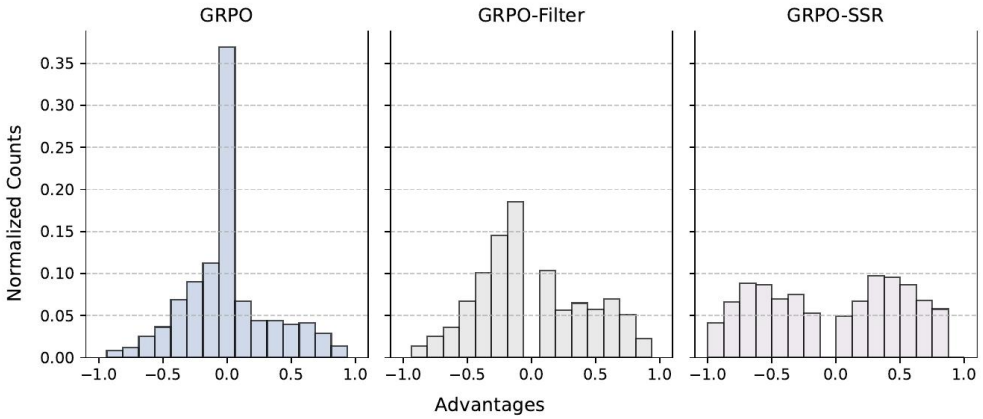

Figure 5: **Advantage distributions during training.** GRPO-SSR maintains a richer, non-zero advantage distribution throughout training, compared to the zero-dominated distributions of GRPO and GRPO-Filter.

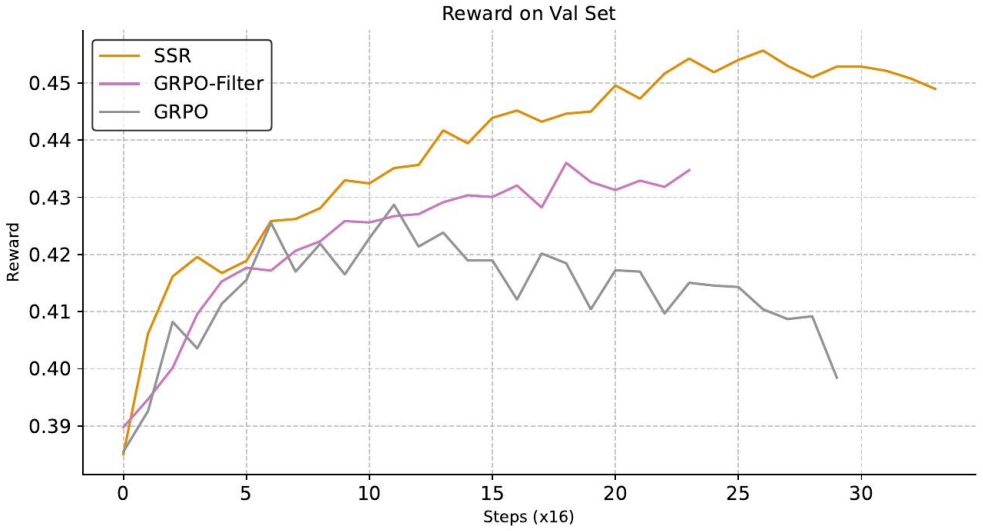

Figure 6: **Evaluation accuracy curves.** GRPO-SSR achieves higher accuracy and more stable improvement compared to vanilla GRPO and the filtering baseline, demonstrating the benefit of sustained learning signal from the replay buffer.

row) reduces MathVision from 32.3 to 29.1 and MathVista from 74.9 to 73.5. Further removing science diagrams ("no Science&Text", 11K samples) drops MathVision to 28.0 and MMMU-Pro to 39.7. These results confirm that all three data types contribute complementary training signal, and that text-only math queries help preserve and generalize reasoning capabilities across benchmark categories.

Rethinking Strategies. Figure 7 analyzes different rethinking trigger strategies and the effect of the fraction q. We find that a moderate rethinking fraction (q=0.5) achieves the best results: too-aggressive rethinking (q=1.0) can disrupt fluent reasoning chains, while too-little (q=0.1) fails to adequately promote self-verification behaviors. Using the set of four trigger phrases with dynamic selection (our default) also outperforms using only a single fixed trigger, suggesting that variety in rethinking prompts helps the model develop more robust and diverse verification strategies.

Model	RL-Algo	Data	MathVision	MathVista	MathVerse	MMMU-Pro	EMMA
VL-Rethinker-7B	SSR	16K	32.3	74.9	54.2	41.7	29.7
w/o 'Forced-Rethinking'	SSR	16K	29.8	72.4	53.2	40.9	29.5
- no SSR	Filter	16K	28.5	72.0	50.0	40.0	26.9
- no SSR& Filter	GRPO	16K	26.0	70.9	51.4	38.8	26.2
- no Text	SSR	13K	29.1	73.5	53.5	41.1	28.7
- no Science&Text	SSR	11K	28.0	71.6	50.3	39.7	28.0

Table 3: Ablation Results to show the impact of SSR and Data Mix.

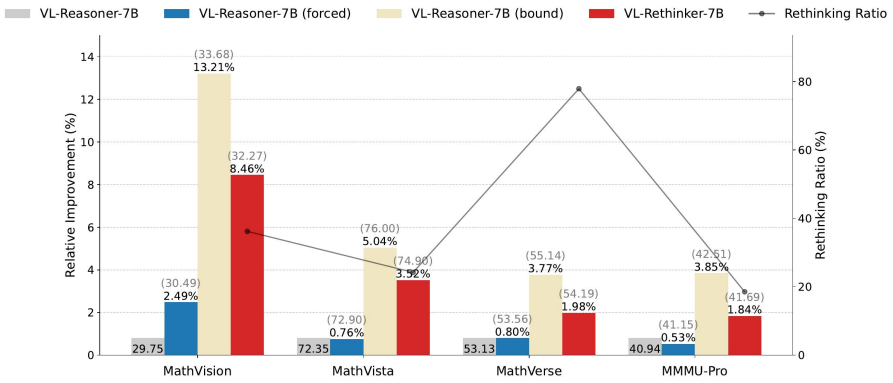

Figure 7: **Relative Improvement with Different Re-thinking Strategies.** Comparing various trigger configurations and forced rethinking variants. Dynamic multi-trigger selection achieves the best overall performance.

5 Related Work

Vision-Language Models. The rapid development of VLMs has produced a family of increasingly capable models for multimodal understanding. LLaVA Liu et al. [2023] and its successors Li et al. [2024], Xu et al. [2024] pioneered visual instruction tuning by projecting visual features into the LLM input space. More recent models such as Qwen2.5-VL Bai et al. [2025], InternVL2.5 Chen et al. [2024a], and DeepSeek-VL2 Wu et al. [2024] have scaled both model capacity and training data, achieving strong performance on diverse benchmarks. Despite these advances, most VLMs rely primarily on supervised fine-tuning and lack explicit mechanisms for multi-step reasoning.

Reinforcement Learning for Multimodal Reasoning. RL-based training has emerged as a key paradigm for inducing reasoning in LLMs. OpenAI o1 Jaech et al. [2024] and DeepSeek-R1 Guo et al. [2025] demonstrated that RL with chain-of-thought Wei et al. [2022] supervision can produce models capable of extended deliberation. Subsequent work has explored RL for reasoning in VLMs: Vision-R1 Huang et al. [2025] and LMM-R1 Peng et al. [2025] apply GRPO to visual reasoning tasks, while R1-OneVision Yang et al. [2025] and OpenVLThinker Deng et al. [2025] scale this approach with curated datasets including distilled reasoning traces. MM-Eureka Meng et al. [2025] and R1-VL Zhang et al. [2025] further explore reward design and step-wise training strategies. SimpleRL Zeng et al. [2025] shows that simple RL recipes can be surprisingly effective across model families. However, none of these methods explicitly address the vanishing advantages problem that limits GRPO effectiveness, nor do they propose mechanisms to train self-reflection behaviors without teacher distillation.

Improvements to GRPO and RL Stability. Several concurrent works have proposed modifications to GRPO for text-domain reasoning. Dr. GRPO Liu et al. [2025] identifies and removes length bias in GRPO, improving token efficiency. DAPO Yu et al. [2025] introduces decoupled clipping, dynamic sampling, and token-level loss for large-scale RL training. Light-R1 Wen et al. [2025] and Open-Reasoner-Zero Hu et al. [2025] explore curriculum-based and simplified training recipes. Our work is complementary to these efforts and specifically targets VLM-specific challenges: the multimodal distribution shift and the advantage sparsity arising from binary rewards on visual reasoning.

Prioritized Replay and Curriculum Learning. Prioritized experience replay (PER) Schaul et al. [2015] selects transitions based on temporal-difference error magnitude, improving sample efficiency in deep RL. Curriculum learning Wang et al. [2022] organizes training data from easy to hard, facilitating smoother optimization. SSR draws on both ideas but operates at the GRPO group level, selectively replaying samples with informative advantages rather than individual transitions or fixed difficulty orderings.

Self-Reflection and Rethinking. Process reward models Lightman et al. [2023] and tree-search methods have been used to train step-level self-correction. Budget forcing Muennighoff et al. [2025] appends "Wait" tokens at test time to extend thinking. Unlike these approaches, Forced Rethinking is a *training-time* signal that the model internalizes, teaching it to self-verify without test-time intervention. Virgo Du et al. [2025] and similar works show that even simple textual cues can trigger reasoning improvements, motivating our choice of short, natural trigger phrases.

6 Conclusion

We presented `VL-Rethinker`, a reinforcement learning framework for training self-reflective Vision-Language Models. By identifying the vanishing advantages problem in GRPO-based VLM training and addressing it with Selective Sample Replay and Forced Rethinking, `VL-Rethinker` achieves state-of-the-art performance across six multimodal reasoning benchmarks at both 7B and 72B scales. Our results demonstrate that effective multimodal slow-thinking can be cultivated through stable RL training, without requiring distillation from proprietary teacher models.

Key findings include: (1) the vanishing advantages problem is a critical bottleneck in VLM RL training, accounting for up to 6.3 points of degradation when unaddressed; (2) SSR alone provides large gains by maintaining a healthy gradient signal throughout training; (3) Forced Rethinking further improves performance by explicitly training self-verification behaviors; and (4) diverse training data including text-only math is important for generalization.

Future work may explore extending `VL-Rethinker` to additional modalities such as video and audio, applying SSR to other RL algorithms beyond GRPO, and scaling to even larger model sizes. We also plan to investigate more sophisticated rethinking triggers and adaptive selection strategies.

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